How-To Library

Informatica

Upgrading Secure@Source from Version 3.0 to 4.0

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Abstract

This article describes how to upgrade Secure@Source from version 3.0 to version 4.0.

Supported Versions

- Secure@Source 3.0 configured on an embedded Hadoop cluster.

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Before You Begin

Perform the following tasks before you start the upgrade:

General Configuration

1. Verify that Secure@Source 3.0 is up and running with the following services:
   - Model Repository Service
   - Content Management Service
   - Data Integration Service
   - Informatica Cluster Service
   - Catalog Service
   - Vibe Data Stream Service (optional)
   - Secure@Source Service
2. Log into the Secure@Source UI and verify that none of the Jobs are in Failed, Paused, or Running state. Jobs must be in either Complete or Terminated state.

Installer

Download the Secure@Source 4.0 installer released by Informatica and place it in a directory. For example: `<Secure@Source_4.0_Installer>/tmp`.

EBFs

Download the following Emergency Bug Fixes (EBFs) released by Informatica:

- EBF-10229: Enterprise Information Catalog for Informatica 10.1.1.
- EBF-10119: Secure@Source upgrade.
- EBF-10614: Secure@Source consolidated fixes and performance.
- EBF-10619: Certificate issue resolution.

Steps to Upgrade Secure@Source from Version 3.0 to 4.0

Perform the following tasks to upgrade an instance of Secure@Source that is deployed on an embedded Hadoop cluster.

1. From the Administrator tool, disable the following services:
   - Catalog Service
   - Content Management Service
   - Data Integration Service
   - Model Repository Service
   - Secure at Source Service
2. Apply the Enterprise Information Catalog **EBF-10229**.
3. Back up the catalog using the backupContents command.
   a. Set the INFA_TRUSTSTORE and INFA_TRUSTSTORE_PASSWORD variables at the current session to run the backup command.
      - `export INFA_TRUSTSTORE_PASSWORD='<encrypted_truststore_password>`

        To get the encrypted password, add `$INFA_HOME/server/bin` to the PATH variable and then run the following command from the`$INFA_HOME/server/bin` folder:

        ```
        ./pmpasswd pass2038@infaSSL -e CRYPT_SYSTEM
        ```
      - `export INFA_TRUSTSTORE=$INFA_HOME/services/shared/security`

   b. Edit the `infacmd.sh` file in `$INFA_HOME/isp/bin` folder and add the following value after `${JAVA_HOME}/bin/java`:

      ```
      -Djavax.net.ssl.trustStore=<INFA_HOME>/services/shared/security/infa_truststore.jks
      -Djavax.net.ssl.trustStorePassword=pass2038@infaSSL
      ```

   c. Create a folder for the backup files.
   d. Run the following command from `$INFA_HOME/isp/bin` to backup the catalog:

      ```
      sh infacmd.sh ldm backupContents -dn <Domain_name> -un
      <Domain_username> -pd <Domain_password> -sn
      <Catalog_service_name> -of
      <backup_file_including_path>
      ```

      Example:

      ```
      sh infacmd.sh ldm backupContents -dn Domain -un Administrator
      -pd Administrator -sn Catalog_Service -of
      /data1/ilmsatsqe/Backup/Informatca.zip
      ```
4. From the Administrator tool, disable the Informatica Cluster Service.
5. Note the following details from the Administrator tool:
   - Name of the Secure@Source Service.
   - JDBC URL of the Secure@Source repository database.
   - User name and password of the Secure@Source repository database.
6. From the Administrator tool, detach the Vibe Data Stream Service if it is attached to the Secure@Source Service.
7. If present, delete the Vibe Data Stream Service on the Administrator tool.
8. Delete the Secure@Source Service from the Administrator tool.
9. If the admind and node are running, stop both.
10. Shut down the domain and unset the INFA_HOME variable if set.

    ```
    unset INFA_HOME
    ```

11. Open a new putty session.
12. Extract the files from the Secure@Source installer zip file.
13. Download the Ambari binaries and scanner binaries to the `<extracted_dir>/Server/LDM/source` folder.
14. Go to the `<extracted_dir>/Server/LDM` folder.
15. Run the `install.sh` file and enter the following values to upgrade Secure@Source.
    a. On the **Welcome** panel, enter **2** to select the **Upgrade Informatica** option.
    b. Enter **2** to select the **Run the Informatica services upgrade installation** option.
    c. On the **Welcome - Step 1 of 9** panel, enter **2** to agree to the terms and conditions.
    d. On the **Component Selection - Step 1A of 9** panel, enter **2** to select the **Upgrade Informatica Enterprise Information Catalog** option.
    e. Enter **2** to indicate **Yes** to the **Install Enterprise Information Catalog on an embedded Hadoop cluster managed by Informatica?** question.
    f. Enter **2** to indicate that you have read and accept the terms and conditions for the Java SE Development Kit Software.
    g. On the **Upgrade Prerequisites - Step 2 of 9** panel, verify the prerequisites.
    h. On the **Upgrade Directory - Step 3 of 9** panel, enter the directory of the 3.0 Secure@Source instance that you want to upgrade.
    i. Enter the directory for the 4.0 Secure@Source directory.
    j. Enter the domain user name and password.
    k. Continue selecting options and specifying values in the installer until completion.

    Note: If the warning *Stop all Informatica services and domain before proceeding.* appears, ignore it.
16. After the Secure@Source installer completes the upgrade, bring down the domain by going to the new INFA_HOME path.
    a. Go to the `INFA_HOME/tomcat/bin` folder.
    b. Run `sh infaservice.sh shutdown`
17. Copy the `Default.keystore` file from the `$INFA_HOME/tomcat/conf/Default.keystore` folder for the previous version to the `$INFA_HOME/tomcat/conf/Default.keystore` folder for the upgraded version.
18. Copy the contents from the `$INFA_HOME/services/shared/security/` folder for the previous version to the `$INFA_HOME/services/shared/security/` folder for the upgraded version.
19. If any changes were made to the `odbc.ini` and `scannerDeployer.xml` files, copy both files from the folder for the previous version to the folder for the upgraded version.

    To locate the `scannerDeployer.xml` file, use the following command:

    ```
    ls $INFA_HOME/services/CatalogService/ScannerBinaries/CustomDeployer
    ```
20. Verify that the path is updated to the current INFA_HOME for all driver paths in the `odbc.ini` file.
21. Copy any third party libraries such as `JavaPasswordSDK.jar` or any other source related jars like Teradata from the Secure@Source 3.0 `$INFA_HOME/java/jre/lib/ext/` folder to the same folder for Secure@Source 4.0.
22. Go to the `<extracted_dir>/Server/SATS` folder and run the `install.sh` and install Secure@Source.
23. Start the Informatica domain from the upgraded directory.

24. From the Administrator tool, update the keystore or truststore location on the Process tab for each of the applicable services:
    - Catalog Service
    - Content Management Service
    - Data Integration Service
    - Informatica Cluster Service
    - Secure@Source Service
25. Apply the Enterprise Information Catalog **EBF-10619**.
26. Enable and upgrade the following services:

    To enable, select the service in the Administrator tool and click **Actions** > **Enable Service**.

    To upgrade, select the service in the Administrator tool and click **Actions** > **Upgrade**.

    a. Model Repository Service

       Use the following command if you do not want to upgrade from the Administrator tool:

       ```
       <INFA_HOME>/isp/bin/infacmd.sh mrs upgradeContents -dn <Domain> -un Administrator -
       pd Administrator -sn <MRSSERVICE>
       ```

    b. Data Integration Service

       Note: If the Hadoop Distribution Directory property was updated in 3.0 then update the property before enabling the service. Select the following upgrade options from the Actions menu:

       - Upgrade Core Service
       - Upgrade Profiling Warehouse Database Contents

    c. Content Management Service

       Enable the service. You do not need to upgrade.
27. Enable the Informatica Cluster Service.

    To enable, select the service in the Administrator tool and click **Actions** > **Enable Service**.
28. Upgrade the Informatica Cluster Service.

    If you use the Administrator tool to perform the upgrade, the Informatica Cluster Service is upgraded automatically.

    When you upgrade from the Administrator tool, you must provide the full path and file name of the backup file.

    Note: If the Informatica Cluster Service was enabled for SSL before the upgrade, then SSL is enabled for the Ambari server after the upgrade.
29. The upgrade process deletes the data from the cluster during the upgrade and does not restore the data. For this reason, you must restore the backed up catalog data manually after you complete the upgrade.

    a. Verify that the back up zip file is valid.

       The upgrade process does not validate the backup content, but verifies that a backup file is provided in `.zip` file format.

    b. Copy the zip file to the Hadoop machine.

    c. Extract the zip file and place the extracted contents in the `/Informatica/LDM` folder on the Hadoop Distributed File System.

    d. Log in as the HDFS user and run the following command to place the backup file in the Catalog:

       ```
       hdfs dfs -put <extracted_backup_folder_Domain_..> /Informatica/LDM
       ```
30. Enable the Catalog Service.

To enable, select the service in the Administrator tool and click **Actions** > **Enable Service**.

31. Upgrade the Catalog Service.

    After you start the Catalog Service, the Administrator tool prompts you to upgrade the Catalog Service.

    a. To upgrade the Catalog Service, click **Actions** > **Upgrade**.

       Note: After upgrading the Catalog Service, Enterprise Information Catalog creates the catalog content. You must not use Enterprise Information Catalog until the catalog content is created and indexed.

    b. Add the following property in the Catalog Service **Custom** property:

```
LdmCustomOptions.solrdeployment.solr.options=-
Dsolrconfig.updatehandler.autocommit.maxdocs=10000
-Dsolr.autoSoftCommit.maxTime=900000
```

    c. Restart the Catalog Service.

    d. Select **Actions** > **Re-index Content** to manually index the content.

    e. Remove the following property after the re-index completes.

```
LdmCustomOptions.solrdeployment.solr.options=-
Dsolrconfig.updatehandler.autocommit.maxdocs=10000
-Dsolr.autoSoftCommit.maxTime=900000
```

    f. Recycle the Catalog Service.

    Wait until the content reindexing is complete before using Enterprise Information Catalog.

32. Create the Secure@Source Service with the details noted in step 5.

    a. Select the **Content already exists** option.

    b. Clear the **Enable Service** option.

    Note: Do not upgrade the contents at this point.

33. Apply the Secure@Source **EBF-10119**.

    a. Update the Secure@Source repository. Run the following command on the Secure@Source repository:
    `UPDATE SATS_REPO_ATTR_VALUE SET ATTR_ID=117 WHERE ATTR_ID=66`

34. Delete the folder `contexthelp_en_US` from `<INFA_HOME>/services/SecureAtSourceService/`

35. Restart the domain.

36. Upgrade the Secure@Source Service contents.

37. Start the Secure@Source Service.

38. From the Secure@Source UI Jobs page, System Jobs tab, verify that the upgrade job is running.

    Note: Do not go to any other Secure@Source UI page until the upgrade job completes.

39. Apply the Secure@Source **EBF-10614**.

40. Reassign Secure@Source permissions and roles.

After you Upgrade

If you plan to scan Hive, HDFS, or Microsoft SQL Server sources, you must complete additional steps after the upgrade.

Hive and HDFS Data Stores

When you upgrade to 4.0, the Hive or HDFS data stores you created in Secure@Source 3.0 appear in the Not Complete state. Perform the following tasks to restore the data stores to the Complete state.

1. Verify that you have the cluster configuration object in the Administrator tool.

2. Make a note of the cluster configuration ID.
3. Log into the Secure@Source UI.
4. On the **Data Stores** workspace, edit each Hive and HDFS data store.
   a. Add the cluster configuration ID to the **Cluster Configuration ID** property on the **Edit <data store name>** page.
   b. Specify the value for all other required properties.

      For definitions of data store properties, see the *Informatica Secure@Source 4.0 User Guide*.
   c. Click **Save**.

   The Hive and HDFS data stores now appear in the Complete state and are ready to be scanned.

Microsoft SQL Server Data Stores

With the Secure@Source 4.0 upgrade, you no longer have to configure the ODBCINI environment variable. Also, you do not have to add the odbc entry in the odbcini file. Instead, perform the following steps before you scan a Microsoft SQL Server data store:

1. Determine if the ODBCINST environment variable is set to the directory that contains the `odbcinst.ini` file.

   If the ODBCINST environment variable is not set to the directory, complete the following steps:
   a. Set the ODBCINST environment variable to the directory that contains the `odbcinst.ini` file.

      For example:

      ```
      export ODBCINST=$INFA_HOME/ODBC7.1/odbcinst.ini
      ```

      or

      ```
      export ODBCINST=$ODBCHOME/odbcinst.ini
      ```
   b. Restart the Informatica domain.
2. Change the **Connection String** property of the Microsoft SQL Server data stores to: `<host>,<port>@<database_name>`

   For Named Instance, change to: `<host_name\instance_name>@<database_name>`

   You can now scan the Microsoft SQL Server data stores.

Troubleshooting

If you encounter a problem while performing the upgrade, review the following list of known issues and their solutions.

1. Upgrade fails because the `$INFA_HOME/SecureAtSourceService/lib` folder has multiple upgrade and repository jars.

Complete the following steps to resolve the issue:

1. Ensure that the `$INFA_HOME/SecureAtSourceService/lib` folder contains the following jar files:
   - `com.infa.sats.content.repo-10.2.0.1836.490-SNAPSHOT.jar`
   - `com.infa.schema.upgrade-10.2.0.1836.490-SNAPSHOT.jar`
2. Delete any other versions of the jar files listed in the previous step.
3. Go to the Administrator tool and resume the upgrade process from Step 36: Upgrade the Secure@Source contents.

2. Upgrade fails with the following error: `[informatica][Oracle JDBC Driver][Oracle]ORA-01407: cannot update ("INFA_SATS"."SATS_FIELD_DOMAIN"."OBJECT_ID") to NULL; nested exception is java.sql.SQLException:`

`[informatica][Oracle JDBC Driver][Oracle]ORA-01407: cannot update ("INFA_SATS"."SATS_FIELD_DOMAIN"."OBJECT_ID") to NULL`

This issue occurs because Secure@Source removes the data store ID of deleted data stores from the SATS_COLUMN_AUGMENT table but fails to remove the data store entry from the SATS_COLUMN_DOMAIN table.

Complete the following steps to resolve the issue:

1. Delete the extraneous data store IDs from the SATS_COLUMN_DOMAIN table with the following command:

```
DELETE FROM SATS_COLUMN_DOMAIN WHERE COLUMN_ID NOT IN (SELECT COLUMN_ID FROM SATS_COLUMN_AUGMENT)
```

2. Go to the Administrator tool and resume the upgrade process from Step 36: Upgrade the Secure@Source contents.

3. Upgrade fails with the following error: `Error Creating Contents java.sql.SQLException: [informatica][Oracle JDBC Driver][Oracle]ORA-54032: column to be renamed is used in a virtual column expression.`

Complete the following steps to resolve the issue:

1. Find the tables that have the virtual or hidden columns. Use the following command:

```
SELECT TABLE_NAME, COLUMN_NAME, DATA_DEFAULT, HIDDEN_COLUMN
FROM DBA_TAB_COLS c
WHERE c.VIRTUAL_COLUMN = 'YES' AND
c.OWNER  IN ('SATS_REPO_NAME')
```

2. Use the output from the previous step to run the following script against the Secure@Source repository:

```
BEGIN
DBMS_STATS.drop_extended_stats('REPO_NAME', 'TABLE_NAME',
'("COLUMNNAME1","COLUMNNAME2")');
END;
```

Example:

```
BEGIN
DBMS_STATS.drop_extended_stats('INFA_CompanyName_SATS', 'SATS_DATA_DOMAIN',
'("DATA_DOMAIN_ID","DATA_DOMAIN_NAME")');
DBMS_STATS.drop_extended_stats('INFA_CompanyName_SATS', 'SATS_HIST_COLUMN_AUGMENT',
'("SCAN_RUN_ID","IS_SENSITIVE")');
DBMS_STATS.drop_extended_stats('INFA_CompanyName_SATS', 'SATS_HIST_COLUMN_DOMAIN',
'("SCAN_RUN_ID","IS_SENSITIVE")');
DBMS_STATS.drop_extended_stats('INFA_CompanyName_SATS', 'SATS_HIST_COLUMN_DOMAIN',
'("COLUMN_ID","SCAN_RUN_ID")');
DBMS_STATS.drop_extended_stats('INFA_CompanyName_SATS',
'SATS_JOB','("JOB_ID","DEFN_ID","REPO_ID","SCAN_RUN_ID","NAME","DESCRIPTION","STATUS"
,"CREATED_BY","START_TIME","END_TIME")');
END;
```

3. Go to the Administrator tool and resume the upgrade process from Step 36: Upgrade the Secure@Source contents.

4. The Upgrade job fails. with the following error: `ERROR [JobStepHandler] [JOB_TYPE=SYSTEM:JOB_ID=118:JOB_STEP_ID=565]--[SYSTEM] An error occurred while running the job step with ID 565.com.informatica.ilm.sats.services.job.SatsJobException: An error occurred while running the job step with ID 565.Catalog Service Error :JSONArray[-1] not found. com.informatica.ilm.sats.services.job.SatsJobException: An error occurred while running the job step with ID 565.com.informatica.ilm.sats.services.job.SatsJobException: An error occurred while running the job step with ID 565.Catalog Service Error :JSONArray[-1] not found.`

Complete the following steps to resolve the issue:

1. Sync the users and locations from the Enterprise Information Catalog UI.
2. Resume the Upgrade job.

5. Cannot enable a service because the keystore certificate has expired.

To resolve, generate the keystore with valid certificates. Then update the information in all the services and restart.

For information on updating keystore and truststore certificates, see the *How to Update Keystore and Truststore Certificates to Maintain Secure Communication* H2L article at https://kb.informatica.com.

6. When you upgrade to Secure@Source 4.0 and scan a Hive data store for the first time, an error appears if you previously scanned the same data store with Secure@Source 3.0.

The following error appears:

```
<Timestamp> - INFO  [testConnection_1515438853160_HiveScanner:MIMBUtilityLogger@26] -
ERROR MIMB_E0042 [BLIB_F0115] Unable to open '/Data/Informatica/SATS4/services/
CatalogService/scanner_miti/Hive/hortonworks_2.5/Drivers/lib'. The error is: 'Path does
not exist'

<Timestamp> - INFO  [testConnection_1515438853160_HiveScanner:MIMBUtilityLogger@32] -
MITI.MIRException: [BLIB_F0115] Unable to open '/Data/Informatica/SATS4/services/
CatalogService/scanner_miti/Hive/hortonworks_2.5/Drivers/lib'. The error is: 'Path does
not exist'

at
MITI.bridges.hive.Import.service.HiveImportBridgeInterface.appendUserClassPath(HiveImport
BridgeInterface.java:138)
at
MITI.bridges.hive.Import.service.HiveImportBridgeInterface.configureClassPath(HiveImportB
ridgeInterface.java:159)
at
MITI.bridges.hive.Import.service.HiveImportBridgeInterface.createJavaBridge(HiveImportBri
dgeInterface.java:75)
at
MITI.bridges.hive.Import.service.HiveImportBridgeInterface.test(HiveImportBridgeInterface
.java:97)
at MITI.mimb.executable.actions.TestAction.run(TestAction.java:161)
```

Complete the following steps to resolve the issue:

1. Disable the Catalog Service.
2. Open the `scannerDeployer.xml` file.
   You can locate the file in the following directory: `$INFA_HOME/services/CatalogService/ScannerBinaries/CustomDeployer`
3. Remove the following lines from the file:

```
<ExecutionContextProperty isLocationProperty="true"
dependencyToUnpack="hive_hortonworks_2.5.zip">
     <PropertyName>HiveScanner_Hortonworks_DriverLocation</PropertyName>
     <PropertyValue>scanner_miti/Hive/hortonworks_2.5/Drivers</PropertyValue>
</ExecutionContextProperty>
```

4. Save the file.
5. Start the Catalog Services.

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